



U.S. National Science Foundation

Birth of the Commercial Internet

The internet as we know it today began with government-funded networking efforts, including NSF's NSFNET.



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The U.S. National Science Foundation catalyzed the creation of the commercial internet that we know today, drawing on the U.S. Department of Defense's pioneering support for early computer networking projects and internet protocols.



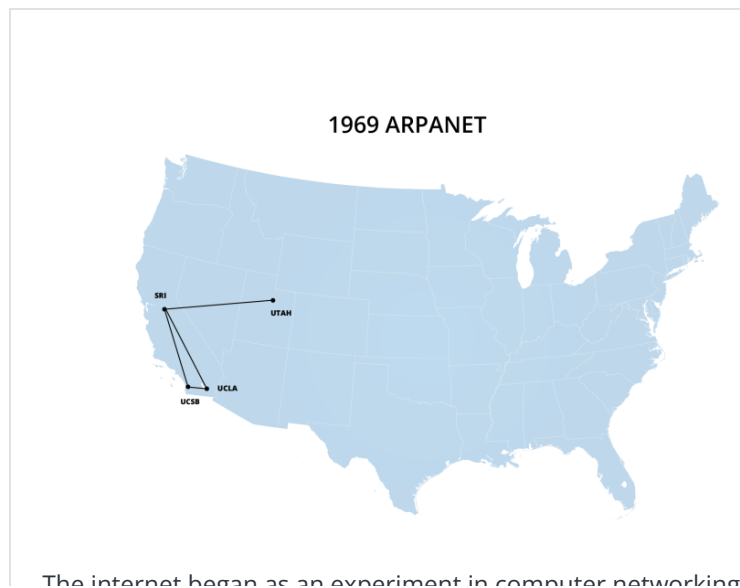
Two computers that sent and received the first email message using the ARPANET.
Source: Dan Murphy

ARPANET: Paving the way

Between 1962 and 1963, Joseph Carl Robnett Licklider wrote a series of memos detailing the earliest ideas and challenges behind establishing "Intergalactic Computer Network" where anyone could quickly access data and information on every subject imaginable. This would ultimately lead to the creation of the internet's direct precursor — the Advanced Research Projects Agency Network (ARPANET) — run by the Defense Advanced Research Projects Agency (DARPA).

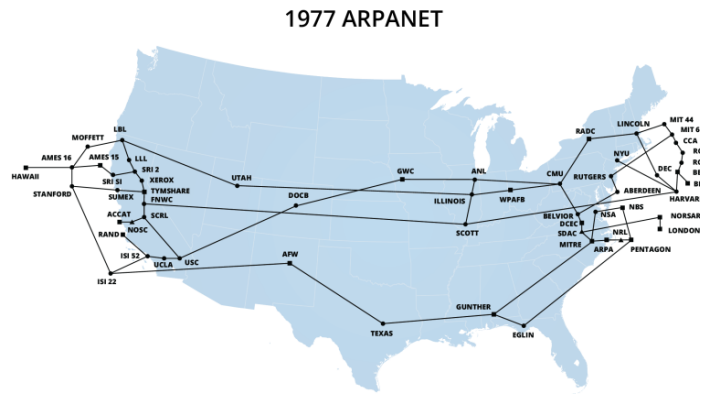
In its earliest form, ARPANET began with four computer nodes in late 1969. Over the next two decades, DARPA-funded researchers expanded ARPANET and designed internet protocols — like transmission control protocol/internet protocol (TCP/IP) — allowing computers to transmit and receive data such as email, computer-to-computer file transfer and remote login.

Learn more about ARPANET



The internet began as an experiment in computer networking by the Department of Defense in the late 1960s.

Credit: U.S. National Science Foundation



After establishing ARPANET in the 1960s, the Department of Defense spent the next two decades expanding its network to include several research universities and laboratories.

Credit: U.S. National Science Foundation

NSFNET: The backbone of the early internet

One of the most significant TCP/IP-based networks was NSFNET, launched in 1986 by NSF to connect academic researchers to a new system of supercomputer centers.

As the first network available to every researcher, NSFNET became the de facto U.S. internet backbone, connecting around 2,000 computers in 1986 and expanding to more than 2 million by 1993.

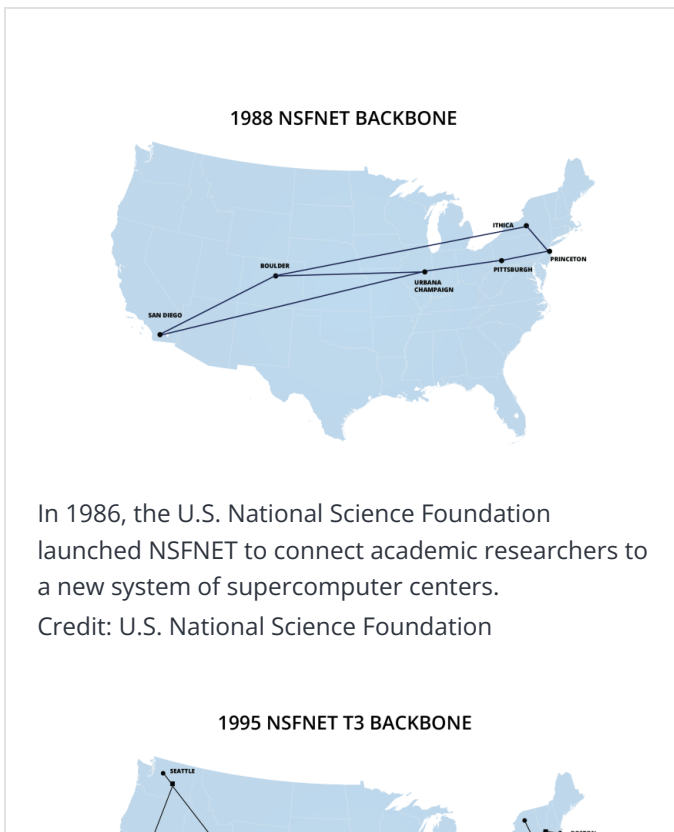
NSFNET laid the foundation for the internet's explosive worldwide growth in the 1990s.

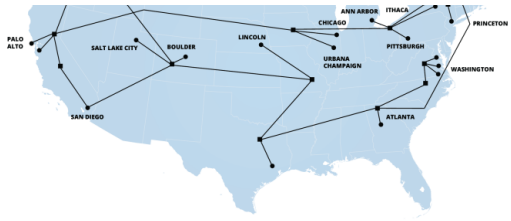


A visualization of inbound traffic measured in billions of bytes on the NSFNET T1 backbone for September 1991. To handle its increasing data traffic, NSFNET became the first national 45-megabits-per-second internet network.

Credit: Donna Cox and Robert Patterson / National Center for Supercomputing Applications and the Board of Trustees of the University of Illinois

Learn more about NSFNET





As the first network available to every researcher, NSFNET became the de facto U.S. internet backbone, connecting around 2,000 computers in 1986 and expanding to more than 2 million by 1993. Rapid expansion of commercial internet services in the early 1990s prompted the U.S. National Science Foundation to shut down NSFNET in 1995.

Credit: U.S. National Science Foundation



ic was the world's first freely available browser.

:: Charles Severance

Going public

Commercial firms noted the popularity and effectiveness of the growing internet and began to build their own network infrastructure, eventually producing products that provided basic connectivity and internet services.

During the 1990s, NSF helped shape the growth and operation of the modern internet, funding the development of the world's first freely available web browser — Mosaic — and algorithms to reduce internet congestion.

The rapid expansion of commercial internet services prompted NSF to shut down its dedicated infrastructure backbone in 1995.

Mosaic: The world's first freely available web browser

The Mosaic web browser allowed browsers to view webpages that included both graphics and text, spurring a revolution in communications, business, education and entertainment that has had a trillion-dollar impact on the global economy.

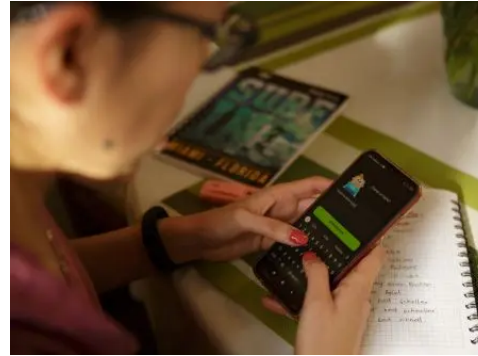
Mosaic was developed out of NSF-funded research at the National Center for Supercomputing Applications at the University of Illinois Urbana-Champaign.

Life after NSFNET

The decommissioning of NSFNET and privatization of the Internet did not mark the end of NSF's involvement in networking.

NSF continues to support many research projects to develop new networking tools, online educational resources and network-based applications.

Over the years, NSF has also been instrumental in providing international connection services that have bridged the U.S. network infrastructure with countries and regions including Europe, Mongolia, Asia, Latin America, Russia and the Pacific Rim.



The Internet has opened up new possibilities for education. For example, the number of people learning a second language with the Duolingo app is more than in the entire U.S. public school system.

Credit: Courtesy of Duolingo